CALL FOR PAPERS

IEEE **WIRELESS COMMUNICATIONS** AND **NETWORKING CONFERENCE**

Pioneering the Future of Wireless Communications

5–8 April 2027 // Panama City, Panama

The IEEE Wireless Communications and Networking Conference (WCNC) is a top-ranked, flagship conference of the IEEE Communications Society, bringing together researchers from academia, industry, and government.

IEEE WCNC 2027 will be hosted in the warm and wonderful city of Panama, and will be conducted in person, allowing attendees to fully benefit from the conference atmosphere and experience.

Prospective authors are invited to submit their works in the form of research papers describing significant and innovative contributions to the field of wireless communications and networking, in accordance with the four technical tracks listed below. Accepted and presented papers will be published in the IEEE WCNC 2027 Conference Proceedings and submitted to IEEE Xplore.

Proposals for half- or full-day tutorials and workshops are also invited in all communication and networking topics.

IMPORTANT DATES

Paper Submission:
15 September 2026

Notification of Acceptance:
15 January 2027

Camera-Ready Papers:
8 February 2027

Workshop Proposal Submission Deadline:
1 July 2026

Tutorials Proposal Submission Deadline:
31 October 2026

To learn more about WCNC 2027 in Panama City and how to submit your paper, please visit our website: wcnc2027.ieee-wcnc.org

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Track 1: Physical Layer and Communication Theory

Track Co-Chairs: Mark Flanagan, University College Dublin, Ireland; Shuowen Zhang, Hong Kong Polythecnic University, China; Ming Zeng, Université de Laval, Canada

- Antennas and RF
- Channel Modeling and Estimation
- Coding Theory and Techniques
- Detection and Estimation Theory
- Energy Harvesting and Low Energy Communication
- Feedback and Two-Way Communication
- Free Space Optical Communication
- Holographic Surfaces and Reconfigurable Intelligent Surfaces
- Information Theory and Electromagnetic Information Theory
- Integrated Sensing and Communications
- Iterative Techniques for Detection and Decoding
- Joint Source-Channel Coding and Cross-Layer Design
- Low-Latency and Short Packet Communications
- Millimeter-Wave and Terahertz
- MIMO, Massive MIMO and Cell-Free Massive MIMO
- Multiple Access and Grant-Free Access
- Near-Field Communication and Sensing
- Physical Layer Security
- Propagation and Interference Modeling
- Reconfigurable Antennas and Fluid Antennas
- Relaying and Self-Backhauling
- Semantic Communications
- Sparse Signal Processing for Wireless Communications
- Waveforms and Modulation
- Wireless Information and Power Transfer

Track 2: Medium Access Control and Networking

Track Co-Chairs: Ramón Agüero, University of Cantabria, Spain; Shuai Han, Harbin Institute of Technology, China; Ayman Radwan, Instituto de Telecomunicações, Portugal

- Age and Value of Information for Networks
- Backscatter Communications
- Cognitive Radio and Networking
- Cooperative Communications and Networking
- Edge Computing, Edge Intelligence, and Fog Networks
- Emerging Medium Access Schemes in the 5G and Beyond
- Energy-Efficient and Green Networking
- Load Balancing and Cell/Band Association
- IoT Networks and Protocols
- Low-Power Wireless Networks
- Multiple Access and Contention
- Multihop Networks
- Network Economics
- Network Slicing
- ORAN Programmability of MAC and Network Functions
- RAN Data Collection and Storage Enhancement
- Resource Allocation for Wireless Communications and Networks
- Resource Management
- Resource Orchestration for Positioning, Navigation, and Timing Systems
- Routing and Congestion Control
- Scheduling and Opportunistic Communications
- SDN/NFV
- Spectrum Sensing, Access, and Sharing
- Unlicensed Spectrum and Licensed/Unlicensed Inter-Networking
- URLLC, Time Sensitive, and Deterministic Networking
- Wireless Network Security and Privacy

Track 3: Machine Learning and Optimization for Wireless Systems

Track Co-Chairs: Sinem Coleri, Koc University, Türkiye; M. Carmen Aguayo, Universidad de Málaga, Spain; Hina Tabassum, York University, Canada

- Bayesian Optimization for Wireless Communications
- Communication-inspired Machine Learning
- Convex and Non-Convex Optimization for Wireless Communications
- Data-driven Network Modelling and Optimization
- Datasets for Wireless Systems and Channels
- Deep Learning for Wireless Communications
- Deep Unfolding for Wireless Communications and Networks
- Distributed Learning and Federated Learning for Wireless Communications
- Distributed Optimization for Wireless Communications
- End-to-end Machine Learning over Wireless Channels
- Game-Theoretic Approaches to Wireless Communications
- Implementation of Machine Learning Algorithms in Wireless Networks
- Large Language Models and Generative AI for Wireless Systems
- Machine Learning Methods for Wireless Localization
- Networking Architectures for Artificial Intelligence
- Online Learning for Wireless Networks
- Performance Analysis of ML Techniques for Wireless Communications
- Reinforcement Learning for Wireless Communications
- Scalability of ML for Wireless Communications
- Semantic and Goal-Oriented Communications
- Transfer Learning for Wireless Communications and Networks
- Unsupervised and Generative Models

Track 4: Emerging Technologies, Network Architectures, and Applications

Track Co-Chairs: M. Fatima Domingues, Khalifa University, UAE; Swades De, Indian Institute of Technology Delhi, India; Lina Zhu, Xidian Univeristy, China

- 5G NR and 6G Standardization
- 802.11 and Next-Generation Wi-Fi
- AI-RAN
- Blockchain and Cryptography
- Connected Vehicles
- Digital Twin Networks
- E-health and Mobile Health
- Experiments, Prototypes, and Testbeds
- Fluid Antenna Communications
- Full-Duplex Communication Networks
- Innovative Implanted and Wearable Devices
- Intelligent Beamforming Relays
- IoT and Machine Type Communications
- Joint Radar and Communications
- Low-altitude Communications and Networks
- Molecular and Nano Communications
- Networking Support for Virtual and Augmented Reality
- O-RAN and Other Compatible Open Radio Access Technologies
- Quantum Communications
- Satellite and Deep Space Communications
- IoT Enablers, Integrated Sensors and Localization
- Software Defined Radio and Networks
- Surface Wave Communications
- UAVs and Non-Terrestrial Networks
- Visible Light and Optical Communication
- Cross-Medium Communication
- Covert Communications

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